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Analysis and optimization of professional soccer player performance using data from activity tracking devices.

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Postgraduate program in Data Science and Machine Learning
Facultad de Ingeniería
Universidad de la República

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Chapter 1

Introduction

1.1. Motivation

In the 2015-2016 european season, FIFA authorized the collection of information from Electronic Performance and Tracking Systems (EPTS) for use during official football competition (Fernández et al., 2016), since then, activity-tracking technologies, including GPS, accelerometers, and other wearable devices, have become increasingly prevalent in professional football worldwide. These tools provide vast data on players' physical performance, offering insights into speed, distance, acceleration, and positioning. However, while data availability has grown considerably, the capacity to promptly extract actionable insights from this data at a minimum cost remains a significant challenge. This gap presents a unique opportunity to combine sports science with data science and machine learning techniques to enhance performance optimization, injury prevention, and tactical decision-making. The motivation behind this thesis arises from the need to bridge the gap between raw data collection and its practical application in the real-time context of professional football, as well as to understand the current status of Uruguayan football in this matter.

1.2. Objectives and Scope

This thesis aims to conduct thorough research on the state of the art in Uruguayan football, focusing on Uruguay's juvenile division performance using a variety of data sources including tracking data, as well as Video processing

and physical status of each player. Over the last few decades, professional football has significantly developed, introducing outside-of-the-box thinking and innovating in every aspect of the discipline by using various devices for taking player metrics, sensors, video analysis, etc. Still, it is well known that these developments haven't fully landed globally in the same way. So one of the objectives is to research what is how Uruguay's situation, determine if it is possible to integrate descriptive and predictive models, and see if there is room to improve decision-making processes for the coaching staff, helping to optimize physical output, anticipate injury risks, and adapt training strategies based on objective evidence.

1.2.1. Research Questions

To guide the development of this thesis, the following research questions have been defined:

- What is data science's state of the art in football, particularly in Uruguay?
- How can machine learning models be applied to predict future performance trends and potential injuries?
- What kind of insights can we obtain based on real-time tracking data support tactical and training-related decisions?
- What are the challenges and limitations of using predictive systems in a professional football environment?

1.2.2. Scope of the Thesis

This research aims to analyze sources of data from professional football players, including data collected during training and matches. The scope includes data processing, feature engineering, exploratory analysis, predictive modeling, and visual monitoring, as well as a dashboard platform development for the coaches and practitioners to follow up and work during the season. The thesis aims not to replace human decision-making but to provide support through data-informed tools. The study is limited to one professional football team with access to tracking devices and historical data.

1.2.3. Structure of the Thesis (REVIEW WHEN FINISH WRITING)

The structure of this thesis is organized as follows:

- **Chapter 2: Background and Related Work** Review existing literature on tracking technologies, sports performance analysis, and machine learning in football.
- **Chapter 3: Methodology** A detailed explanation of the data collection, processing, feature extraction, and modeling techniques used in this study.
- **Chapter 4: Results and Analysis** Presentation of the descriptive and predictive models developed, along with their evaluation metrics and key findings.
- **Chapter 5: Visual Platform and Practical Applications** Description of the developed visual interface and discussion of its practical utility.
- **Chapter 6: Conclusions and Future Work** Summary of contributions, limitations of the research, and potential directions for future studies.

Chapter 2

State of the Art and Theoretical Framework

This chapter will first trace the evolution of performance analysis in football, highlighting how objective recording and examination of behavioral events during competition have become crucial for identifying strengths and weaknesses and for developing sport-specific training protocols (Martins, 2018). It will then detail the increasing role of Electronic Performance and Tracking Systems (EPTS), such as GPS, in collecting vast amounts of granular data on player movement and physical exertion. Following this, the chapter will explore various applications of machine learning algorithms in football analytics, including injury prediction, performance evaluation, and tactical analysis. The discussion will also encompass the inherent challenges in leveraging these technologies, such as data quality, complexity, integration issues, and the need for standardized methodologies. Finally, this review puts the focus on establishing a robust theoretical framework by surveying the current state of the art and identifying gaps, particularly within contexts like Uruguayan youth football, which remains an under-researched area in the available literature.

Performance

Before reviewing the technologies, methods, and data used in football performance analysis and injury prevention, it is essential to understand what *performance* means in a sporting context. This section introduces a theoretical framework that helps organize and interpret the many factors in athletic performance. The purpose of presenting this context now is to understand bet-

ter what is being measured, why it matters, and how different forms of data (like GPS or video tracking) relate to the broader concept of performance in football.

One of the most widely accepted models in this area is the multidimensional model of sports performance proposed by **Morante (2006** [**TODO: LINK AL ARTICULO**](https://altorendimiento.com/analisis-del-rendimiento-en-deportes-de-equipo-del-registro-estadistico-al-analisis-tactico/?srsltid=AfmBOorzZpE3a7-XXMxfU-XkjnShM318bkklnUrwYgTLxXE6dLtynelS)**)**. This framework divides performance into several interrelated dimensions: physical, technical, tactical, psychological, and contextual. Rather than looking at athletes as just physical machines, this model encourages a more complete view, where performance results from many interacting elements. The dimensions of the model are described as follows:

- **Physical Dimension**: This includes speed, endurance, strength, power, and flexibility. These are modern sports' most commonly measured aspects, especially using ETPS technologies.
- **Technical Dimension**: Technical performance refers to the player's ability to execute specific actions correctly and efficiently (e.g., ball control, passing, shooting). These skills are biomechanical and depend on learning, repetition, and adaptation.
- **Tactical Dimension**: Sometimes referred to in the model as "active game actions", this includes decision-making, positioning, reaction to opponents, and other behaviors that take place in the dynamic context of a match. Tracking data (e.g., player movements and positioning) is beneficial for analyzing this area.
- **Psychological Dimension**: This includes motivation, confidence, stress management, attention, and emotional control. Although harder to quantify, these factors directly affect how players train and perform. Some advanced analytics systems now try to include indirect indicators of psychological state (e.g., variability in performance, behavioral patterns under pressure).
- **Contextual Dimension**: Performance does not happen in a vacuum. Weather conditions, the type of pitch, the opponent, match importance, or travel fatigue can all influence how players perform. These external

variables are increasingly considered in injury risk models and workload management systems.

This model is helpful in two main ways. First, it shows that performance is complex and multidimensional. It is not enough to measure how far a player runs or how many accelerations they perform, so the goal is to interpret these values within a larger system that includes context, decision-making, technical execution, and psychological state. Second, it helps to structure the use of modern technologies and data. For example, later sections will discuss how GPS systems collect physical and tactical data, how AI models use these inputs to predict injuries (sometimes including contextual variables), and how computer vision studies technical actions. Referring to Morante's framework helps to clarify which dimensions are being addressed in each case and which are still missing or underdeveloped.

This framework also serves as a reference point when comparing international practices with local realities in Uruguay. If specific dimensions are ignored (e.g., lack of tactical analysis or psychological support), it may limit the impact of performance programs, even if advanced technologies are available.

Finally, this model provides a *conceptual foundation* for the research questions explored in this thesis. Understanding how each dimension contributes to performance, and how data and technology can (or cannot) capture them, will help us define the strengths and limitations of current practices and identify areas for improvement in the application of sports science in football.

2.1. Injury Prevention in Football

2.1.1. Physical and Sport Performance Perspective

Understanding the physical aspects of performance is a critical part of any effort to prevent injuries and improve player output in football. As outlined in the previous section, the **physical dimension** is one of the central pillars of sports performance. It includes measurable attributes such as speed, acceleration, deceleration, muscular strength, endurance, and workload. These variables are not only essential to understand how a player performs, but also serve as key indicators of **injury risk**.

Literature on Performance Data and Injuries

The study of performance in football has undergone a profound transformation over the past century. In its early stages, performance analysis was largely observational and anecdotal, relying heavily on the subjective interpretation of coaches and practitioners. Tactical decisions, training plans, and player evaluations were based primarily on experience, intuition, and limited statistical data such as goals scored or distance covered, often compiled manually.

A pivotal shift occurred in the 1960s and 1970s, with the introduction of systematic notational analysis. Researchers such as Charles Reep began quantifying aspects of match play by coding events manually, marking an early attempt to bring objectivity to football analysis (Reep and Benjamin, 1968). Although rudimentary, these studies revealed surprising patterns, such as the limited number of passes leading to most goals, that challenged prevailing assumptions and laid the groundwork for more data-driven approaches.

The technological revolution of the late 20th century catalyzed further innovation. The 1990s witnessed the introduction of video analysis software, enabling detailed post-match reviews. In recent years, the availability of wearable technologies and tracking systems has enabled researchers and practitioners to collect highly detailed performance data. A significant body of research has shown correlations between physical metrics, especially high-speed running, sprint frequency, accelerations, and cumulative training load, and the incidence of musculoskeletal injuries in football players, as we will be covering in future chapters.

For instance, Malone et al., 2018 demonstrated that rapid increases in workload, particularly high-intensity efforts, were associated with a greater risk of soft-tissue injuries. Similarly, Gabbett et al., 2016 work on the *Acute: Chronic Workload Ratio (ACWR)* emphasized that both undertraining and overtraining can increase injury likelihood, reinforcing the need for balanced and individualized training plans. These findings are consistent with the idea that physical performance metrics can serve not just as output indicators, but as early warning signals for potential breakdowns in the athlete's physiological capacity.

Moreover, modern studies do not treat physical load as a static value, but rather as a dynamic and context-dependent variable that interacts with recovery, match demands, and individual player profiles, aligning with the multidi-

mensional view of performance proposed at the beginning of this chapter.

Risk Factors: Intrinsic and Extrinsic Contributors

Injury prevention requires understanding both **intrinsic** and **extrinsic** risk factors. Intrinsic factors are characteristics inherent to the player that align with Morante's scheme and are more related to the physical dimension. On the other hand, extrinsic factors related to the environment or external conditions could be associated more with the contextual dimension. Hence, the interaction between these variables is complex, and modern monitoring systems aim to detect emerging risks before they result in injuries. Significantly, these variables can change throughout a season, which justifies the need for continuous and adaptive monitoring rather than relying on fixed thresholds or assumptions, and one of the most accepted ways of controlling these factors is through controlled cyclical training sessions.

Cyclication: A Methodology for Performance and Injury Prevention

Many elite teams now apply a **cyclical methodology** for injury prevention; This theory is based on the idea of focusing training on different areas over a set period of time (such as a week or a few days, depending on the schedule), with the overall goal of optimizing performance for key competitions, rather than trying to keep athletes at peak performance throughout the entire season (Fister and Jr., 2019). In this method, the performance data is continuously collected, analyzed, and used to inform interventions. This process can be divided into several blocks:

- **Data Acquisition**: Using GPS, inertial sensors, and heart rate monitors to capture physical performance metrics during training and matches.
- **Analysis**: Comparing data across sessions, weeks, or periods to identify unusual patterns or deviations from expected values.
- **Decision-making**: Based on this analysis, coaching and medical staff adjust workloads, introduce recovery sessions, or tailor individual training plans.
- **Reassessment**: Continuously update the data and review the interventions' effects.

This iterative approach reflects a feedback loop, where each stage informs the

next. It also highlights the importance of collaboration between coaches, sports scientists, medical teams, and players. Data is not an end in itself but a tool for informed decision-making.

Worst Case Scenarios: Identifying the Peak Demands

A critical advancement in recent years has been the introduction of **Worst Case Scenario (WCS)** analysis. WCS refers to the most physically demanding passages of play within a match, typically measured over rolling time windows (e.g., the most intense 1-minute, 3-minute, or 5-minute periods).

Rather than analyzing average workload over an entire game, which can underestimate the stress placed on players, WCS captures the maximum intensity periods representing the real physical limits players must be prepared for. These include bursts of repeated sprints, sharp accelerations and decelerations, and high-intensity movements often observed during transitions or pressing phases.

Studies such as Martín-García et al., 2018 have shown that training sessions often fail to replicate the intensity of these worst-case periods, which can result in players being underprepared for peak demands. This mismatch between training and match intensity is a known risk factor for injury and performance drop.

By identifying these scenarios through data, practitioners can design training drills that simulate game-like intensity, monitor players' responses, and assess readiness, integrating physical data within the larger strategic framework discussed at the beginning of the chapter.

2.2. Engineering Tools for Performance and Injury Prevention

Modern football is no longer just about physical preparation or tactical execution; it has been increasingly influenced by technological tools that allow for intelligent decision-making. This section presents these tools, the role they play, its applications, and challenges, showing how they complement the physical and sport performance perspective previously discussed (2.1).

2.2.1. Electronic Performance and Tracking Systems (EPTS)

The limitations inherent in traditional event data, particularly its inability to comprehensively capture off-ball movements and continuous player activity, underscored the critical need for more advanced tracking technologies. This necessity catalyzed the widespread adoption of Electronic Performance and Tracking Systems (EPTS), which offer a richer, more continuous stream of player movement data. EPTS devices quantify external load parameters, which represent the physical demand imposed by a task or exercise. They offer the ability to collect and analyze distance-driven and positional measures during training sessions and matches. FIFA has also officially approved the use of EPTS devices during competitive matches, further solidifying their role in modern football analysis.

EPTS primarily rely on several core technologies to capture granular player data. **Global Positioning Systems (GPS)** are widely utilized in team sports to monitor players' external loads, including movement patterns, speed, and distance covered. These devices commonly integrate a GPS receiver with other microelectromechanical systems (MEMS) such as accelerometers, gyroscopes, and magnetometers. GPS units typically operate at sampling rates of 10 Hz or higher, providing position, velocity, and distance data, making them perfect for its use in team sports. Studies have affirmed their reliability and validity for various measures, including distances, speeds, and accelerations/decelerations. However, GPS accuracy can be affected by factors such as the number of connected satellites or the presence of high and large buildings in the surrounding area, particularly in stadium environments, which can limit satellite reception.

Local Positioning Systems (LPS) offer an alternative, particularly for indoor environments or stadiums where GPS signal might be compromised. LPS systems, such as Kinexon, utilize radio-technology to transmit time signals to antennae, calculating a player's momentary 2D position based on the time-of-flight (TOF) of ultra-wide-band radio signals. These systems typically operate at higher frequencies (e.g., 20 Hz) and have shown acceptable validity for evaluating locomotor patterns in indoor sports compared to motion capture systems, particularly in the center of the playing field.

These two technologies dedicated devices, could be linked together with

multimedia analysis techniques of Artificial intelligence (AI) called Computer Vision (CV) to extract more metrics or cross validate data (Cossich et al., 2023).

2.2.2. Key Physical Metrics (ANDRES REVIEW NEEDED)

EPTS enable the collection of various key physical metrics, broadly categorized into volume and intensity. **Volume metrics** primarily quantify the total amount of work performed. **Total Distance (TD)** covered by a player during a session or match is a fundamental measure, providing an overall indicator of workload. Elite male outfield players typically cover about 10,000 to 13,500 meters during a game, although specific values can vary depending on factors like playing position and league. **High-Speed Running (HSR)** and **Sprint Distance (SD)** represent distances covered above specific velocity thresholds, indicating high-intensity efforts. HSR thresholds can vary (e.g., $\geq$ 5.5 m/s or 19.8 km/h), with sprint defined as sustained running above an even higher speed (e.g., $\geq$ 7 m/s or 25.2 km/h, lasting at least 1 second). These high-intensity efforts are crucial for understanding the physical demands of modern football and have significantly increased over the last decade.

These metrics are highly relevant for understanding the neuromuscular requirements of the sport, as force is proportional to acceleration. The ability to perform repeated bouts of high-speed running, acceleration, and deceleration has been identified as a key characteristic in elite football. **PlayerLoad (PL)**, or modified vector magnitude, is a composite measure derived from accelerometer data, reflecting the overall mechanical stress accumulated by an athlete. And another related metric is the **High Metabolic Load Distance (HMLD)**, which integrate both speed and acceleration/deceleration efforts to provide a more comprehensive view of high-intensity activity. These metrics provide objective data that, when combined with internal load indicators like heart rate, offer a holistic view of a player's physical output and response to training.

2.2.3. Positional Data

The continuous collection of player positions by EPTS has revolutionized the analysis of both individual and team-level tactical behaviors, moving be-

yond simple aggregated statistics. For individual players, this data allows for the creation of *heatmaps*, which visually represent the frequency or density of a player's presence in different areas of the pitch, offering insights into their typical playing zones and work rate distribution. This detailed positional information helps to identify individual playing styles and areas of influence.

At the *team level*, positional data enables a more sophisticated analysis of tactical shape, organization, and collective behavior. It allows for the automatic identification of team formations (e.g., 4-4-2, 4-3-3), even as player roles might change dynamically during a match. This is crucial because physical characteristics are influenced by playing in different formations. Concepts such as pitch control, which evaluates how players use space, and passing networks, which model player interactions through ball distribution, provide deeper insights into team tactics and fluidity. Furthermore, the analysis of distances and proximity between players (both teammates and opponents) can shed light on collective defensive and offensive strategies, such as pressing intensity or exploiting space. The dynamic and multifactorial nature of football performance necessitates this detailed spatial and temporal data to truly understand team organization and player interactions.

2.2.4. Tactical Interpretation

Beyond individual performance metrics, EPTS data, particularly positional information, enables a sophisticated tactical interpretation of player and team behavior, linking physical outputs to strategic objectives and game contexts.

A player's role and position on the field significantly influence their physical demands. For example, midfielders typically cover greater total distances than defenders or forwards, and wide midfielders and forwards often perform more high-intensity running and accelerations/decelerations. Playing position can exert a greater influence on running demands than general physical fitness alone, suggesting that tactical roles constrain a player's movement patterns. Moreover, the physical characteristics demanded of a player are influenced by the specific tactical formation employed by the team.

Team strategy and game state are powerful modulating factors of physical output. Tactical approaches, such as high pressing or defensive blocks, directly influence player activity. Contextual factors like match outcome (win, loss, draw), score-line, quality of opposition, and match location (home or away) have been shown to impact physical outputs. For instance, teams may cover

significantly longer distances in won matches compared to draws or losses. Furthermore, player unavailability due to injuries can lead to increased high-intensity efforts (e.g., sprints) from available players to compensate. Football's dynamic and unpredictable nature means that collective movements and team organization constantly adapt to both teammates and opponents, requiring an integrated analysis of physical and tactical metrics. High-intensity runs, for example, are strongly associated with critical game situations like defensive transitions, recoveries, and progressions into the final third.

2.2.5. Data Governance

The increasing adoption of EPTS in professional football has led to the implementation of governance frameworks to ensure data quality and standardization. The **FIFA Quality Programme for EPTS** serves this purpose, establishing standards for the accuracy and reliability of tracking devices used in competitive play. This program ensures that systems like STATSports APEX (link), widely used by Football, meet specific technical requirements for monitoring multiple satellite systems and incorporating accelerometers and gyroscopes. This standardization is crucial for making comparisons across studies and teams, as different GPS devices can sample at varying frequencies and use other software algorithms, potentially affecting the accuracy of derived variables. By setting benchmarks and validating technologies, FIFA's program aims to enhance the trustworthiness and practical applicability of EPTS data for performance analysis and decision-making within football [1].

2.3. Artificial Intelligence in Football

2.3.1. Machine Learning Applications in Football Analytics

The preceding section underscored the multifactorial nature of physical performance in football and highlighted that interpreting the extensive Electronic Performance and Tracking Systems (EPTS) data within this complex contextual framework necessitates advanced, automated analytical methods. Machine learning (ML) emerges as a powerful paradigm to address this com-

[1] FIFA quality programme for electronic performance & tracking system manual

plexity, offering sophisticated capabilities for pattern recognition, prediction, and optimization in football analytics (Fister and Jr., 2019; Rein and Memmert, 2016).

2.3.2. Models based on Performance & Injury Risk

Machine learning algorithms are increasingly deployed to model training load and predict injury risk, offering a more nuanced approach than traditional statistical methods (Mateus et al., 2025; Martins, 2018). For instance, ML techniques can effectively predict Rating of Perceived Exertion (RPE), a key measure of internal training load, from objective data such as running distances and speeds captured by GPS devices and accelerometers. This predictive capacity allows for real-time estimation of RPE during training sessions, aiding in more precise load management and future training planning (Carey et al., 2016). Furthermore, ML models, particularly Random Forest Regression, have demonstrated superior ability in predicting heart rate (HR) responses during training drills based on external load data, facilitating the development of a "fitness index" that provides insights into a player's training status without requiring additional invasive tests (Mandorino, Clubb, and Lacome, 2024).

The inherent multifactorial and dynamic nature of sports injuries makes them a particularly suitable problem for machine learning applications (Martins, 2018). ML models are used to predict the risk of non-contact lower body injuries, often resulting from overtraining or undertraining (Pilka et al., 2023). Such models can assess injury risks with remarkable accuracy by analyzing multidimensional data, including a player's psychological state, nutrition, sleep patterns, and injury history, alongside external and internal training loads (Mateus et al., 2025). Subjective wellness reporting, encompassing factors like muscle soreness, stress, fatigue, and sleep quality, provides valuable input for these models, often surpassing objective measures in certain contexts (Hoel, 2023; Martins, 2018). The acute:chronic workload ratio (ACWR), which compares recent training load to a longer-term average, is frequently incorporated into these models as a significant predictor of training-related injuries, with rapid increases in workload generally leading to higher injury risk (Pilka et al., 2023). Conversely, maintaining high chronic training loads and adequate exposure to maximal velocity running has been shown to offer a protective effect against injury (Malone et al., 2018). However, despite promising results, limi-

tations such as data sparsity and challenges in generalizability due to reliance on single-team datasets remain in injury prediction research (Martins, 2018 Hoel, 2023).

2.3.3. Enhancing Tactical Analysis

Machine learning significantly enhances tactical analysis by identifying complex patterns and behaviors that are difficult for human experts to discern (Rein and Memmert, 2016). This includes the automated classification of high-intensity runs (sprints) into contextual categories based on their tactical purposes, overcoming the scalability limitations of manual annotation (Kim et al., 2024). ML can automatically identify team formations and their changes during a match, revealing insights such as teams adopting more defensive formations in away games (Bialkowski et al., 2014). Clustering algorithms are used to group player trajectories and identify recurring attacking sequences (Knauf, Memmert, and Brefeld, 2016; Hirano and Tsumoto, n.d.).

Beyond individual actions, ML enables the analysis of collective movements and coordination tactics, moving beyond traditional pass-based networks to capture broader player interactions (Marcelino et al., 2020). Techniques for real-time quantification of "dangerousness" in specific pitch areas, and visual analysis of pressure, are also facilitated by spatiotemporal tracking data and ML (Link, Lang, and Seidenschwarz, 2016, Buldu et al., 2020). Value-based metrics, such as expected threat (xT) or similar models, are developed to quantify the offensive value of player actions, including predicting and valuing penetrative passes or assessing the effectiveness of pressing (Hassani, Ramdani, and Lotfi, 2025). Integrating physical efforts with tactical phases of play is another critical application, with ML revealing strong associations between high-intensity runs and specific game situations like defensive transitions, recoveries, and progressions into the final third (Bradley, 2024).

2.3.4. Talent Identification note: EXPAND THIS

Machine learning offers significant advancements in talent identification and player recruitment processes (Mateus et al., 2025). By processing vast datasets, ML models can create detailed player profiles that go beyond traditional scouting methods, integrating performance metrics, biometric data, and even simulating potential impact on team performance (Sahin and Erol, 2017; Mccullagh, 2010). This allows clubs to predict a player's future performance

more accurately and understand how a recruit might integrate into existing tactical frameworks (Arndt and Brefeld, 2016). Furthermore, ML can identify key characteristics of individual athletes within the training process, enabling more personalized development strategies (Fister and Jr., 2019). The ability to process extensive data quickly and accurately offers a significant enhancement to player assessment and recruitment, aligning physical characteristics with tactical roles (Bradley, 2024).

2.3.5. Common Algorithms

Several machine learning algorithms are commonly employed in football analytics, broadly categorized into supervised and unsupervised learning approaches (Hoel, 2023).

- **Supervised learning** algorithms rely on labeled data to learn a mapping function from input features to an output variable (Hoel, 2023). Key examples include:
  - **Random Forests (RF)**: Ensemble learning methods used for both regression (e.g., predicting RPE or HR responses) and classification (e.g., injury prediction) tasks, known for handling complex, non-linear relationships (Carey et al., 2016; Mandorino, Clubb, and Lacome, 2024).
  - **Gradient Boosting Machines (GBM) / XGBoost**: Powerful algorithms that build models sequentially, often outperforming other methods in predictive accuracy for tasks like match outcome prediction and injury forecasting (Baboota and Kaur, 2019; Pilka et al., 2023).
  - **Support Vector Machines (SVM)**: Used for classification and regression, particularly effective in finding optimal decision boundaries, applied in match result and injury prediction (Martins, 2018; Leung and Joseph, 2014).
  - **Logistic Regression**: A baseline classification algorithm, commonly used for binary outcomes such as predicting injury occurrence, highlighting linear relationships between variables (Martins, 2018).
  - **Decision Trees**: Simple yet effective models used for classification, for instance, in injury prediction, where they offer interpretability

of influencing factors (Pilka et al., 2023).

- **Neural Networks (NN)**: Algorithms inspired by the human brain, forming the basis of deep learning. They are adept at recognizing patterns and making predictions in various applications, including forecasting match outcomes and identifying player roles (Sahin and Erol, 2017).
- **Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU)**: Specialized neural networks designed for time series data, capable of capturing temporal dependencies and proving highly effective in tasks like injury prediction where sequence data is crucial (Hoel, 2023).
- **ROCKET**: A fast and computationally efficient model that uses convolutional operations for time series classification, demonstrating strong performance in injury prediction (Hoel, 2023).

- **Unsupervised learning** algorithms identify patterns in unlabeled data.

  - **Clustering**: Groups data points based on similarity. In football, this can involve identifying distinct tactical patterns, player movement groups, or performance trends over time for individualized training (Fister and Jr., 2019; Mateus et al., 2025).

The advent of these powerful machine learning models promises to transform football analytics from a descriptive discipline into a predictive and prescriptive one. However, translating these sophisticated models and insights into practical, actionable tools that are seamlessly integrated into the fast-paced environment of professional football clubs presents significant real-world challenges (Rein and Memmert, 2016). These challenges, encompassing data governance, effective communication, and the critical balance between AI-driven insights and human expertise, will be explored in the subsequent section.

Artificial Intelligence refers to the development of computer systems that can perform tasks that typically require human intelligence. In football, AI has become a key instrument for analyzing large volumes of data and extracting insights that are too complex or subtle to detect manually.

One of the most impactful applications of AI in sport is the creation of predictive models that can estimate injury risk before it happens. These models

use historical performance data, physical workload, match intensity, and sometimes even contextual factors (like schedule congestion or weather) to forecast the likelihood of a player sustaining an injury (Mateus et al., 2025).

For example, (Arndt and Brefeld, 2016) explicitly mention the idea that machine learning algorithms can be trained on past data to identify combinations of variables that usually precede an injury (e.g., high cumulative workload combined with low recovery time). Once trained, these models can predict current players' performance based on their recent data, which enables medical staff to take preventive action, such as adjusting training intensity, scheduling recovery sessions, or monitoring specific players more closely.

In (Mateus et al., 2025), the researchers defined deep learning as a more advanced stage of AI, employing neural networks with many layers to process complex data automatically. So, unlike traditional models that use fixed thresholds (e.g., "don't exceed X km per week"), we can use these techniques to learn from patterns in the data, adapting to each player's unique physiology and workload history, resulting in an approach that is more individualized and dynamic.

AI to Enhance Tactical and Physical Performance

Beyond injury prevention, AI is also utilized to optimize tactical performance and manage workload effectively. Some clubs use reinforcement learning or pattern recognition algorithms to study how players move, how teams press, or how passing networks evolve over time. Others apply AI-powered simulations to predict fatigue buildup across a match or season, or recognize patterns that are hard to distinguish (Cossich et al., 2023).

Current Known Applications

Top European clubs and national teams have already adopted AI tools in their performance and medical departments. However, many of these systems are proprietary and not publicly disclosed, which limits academic visibility. Still, research teams from companies like Zone7 have demonstrated that AI models can predict injuries early on and ultimately reduce their incidence, thereby improving decision-making. They share an anonymous outcome of some previous seasons, which are very promising for what we could expect

with a fully allocated team dedicated to that matter.

In August 2025, StatsSports (which is the tool this research is more focused on) released a new module that does some optimizations on AI https://pro.statsports.com/statsports-launch-new-ai-powered-technology/ (**TODO: PREGUNTAR A WILSON**)

It is a common interest....

—

2.3.6. Computer Vision for Data Collection

While AI helps make sense of data, it is useless if there is not enough data to use, or if the quality is not good at all, so this is where Computer Vision comes in, and allows the extraction of visual information from images or videos, and converting it into meaningful data for analysis.

Tracking Player Movements Using Cameras and Algorithms

Traditionally, player tracking has relied on GPS or inertial sensors. However, computer vision-based systems allow for tracking without wearables, using one or multiple synchronized cameras placed around the stadium. These systems use pose estimation, object detection, and optical flow algorithms to detect players' positions, velocities, and tactical team strategies.

Combining Visual and Physical Data for Full Context

The real value emerges when visual tracking data is combined with GPS and biometric data. For example, while a GPS unit can tell you that a player ran 10 km, a vision-based system can reveal how those meters were distributed: how many were covered while pressing, during sprints, or in transitions.

By fusing multiple data sources, we can build a much more complete profile of each player's physical and tactical behavior. This combination is particularly powerful when evaluating worst-case scenarios since it helps us understand certain moments' intensity and biomechanical complexity during a game and tactical strategies.

Challenges and Solutions in Vision-Based Data Capture

Despite its benefits, Computer Vision comes with some challenges:

- **Occlusion**: When players are too close together or blocked from view, algorithms may struggle to detect them correctly.
- **Calibration**: Accurate tracking requires careful camera alignment and calibration, especially in stadiums with variable lighting and layout.
- **Data Volume**: High-resolution video generates massive amounts of data, which must be processed quickly and reliably.

To address these issues, there are some techniques that are currently being utilized to overcome these challenges like use **multi-camera fusion**, **3D reconstruction techniques**, and **machine learning-based corrections** to increase tracking accuracy. Hybrid systems that combine **vision with wearable sensors** are also emerging, allowing for redundancy and error correction.

2.4. Focus on Uruguay and How We Compare Globally

(REVIEW, ANDRES MADE COMMENTS) So far, this chapter has presented a broad and interdisciplinary perspective on performance and injury prevention in football, combining sports science and engineering tools like AI and computer vision. But it's essential to localize this knowledge: understanding what is currently being done in Uruguay, and how it compares to international standards. This allows us to identify both strengths to build on and gaps to address, particularly if we aim to apply these tools in real-world settings or propose new systems tailored to the local context.

2.4.1. Available Programs and Technologies in Uruguay

Uruguay has a strong football tradition, with a long history of international success relative to its population size. However, when it comes to technology adoption and injury prevention systems, the landscape is more heterogeneous.

At the professional level, especially in clubs competing internationally (e.g., Nacional, Peñarol, and the national team), some teams have begun incorporating GPS tracking systems, video analysis platforms, and even basic data science tools for performance monitoring. These usually come from international providers such as Catapult, WIMU, or StatsSports (The one we will be focusing on later), and are used primarily to track distance covered, sprinting volumes, and general workload.

Some medical teams also use subjective wellness questionnaires, neuromuscular testing (e.g., CMJ platforms), and manual video tagging tools to help evaluate physical condition and readiness. However, these tools are often limited by budget constraints, technical staff availability, or lack of integration across departments (e.g., coaches, physiotherapists, data analysts).

At amateur or youth levels, the situation is more fragmented. Most teams rely on coach observations, manual logs, and fitness testing. While there are increasing efforts to introduce technology, access is often restricted to pilot programs or isolated initiatives supported by public institutions or universities.

2.4.2. International Comparison

Compared to elite football environments in Europe, North America, or Asia, Uruguay is still in an early adoption phase. Many top-level clubs abroad use integrated systems that combine AI models, GPS data, computer vision, and medical records in centralized platforms. These systems allow for real-time monitoring, automated risk alerts, and adaptive training plans, all supported by multidisciplinary teams including sports scientists, data analysts, and software engineers.

In countries like Spain, England, or Germany, it's common for clubs to have in-house departments dedicated to data-driven decision making, including professionals trained in machine learning, biomechanics, and software development. These resources are used not only for injury prevention but also for tactical scouting, contract negotiations, and long-term player development.

In contrast, Uruguay faces several structural limitations:

- Smaller budgets and fewer full-time specialists in sports tech.
- Lower access to automated data pipelines and real-time monitoring tools.
- Limited adoption of AI or computer vision systems beyond GPS.
- A stronger reliance on intuition and tradition rather than predictive analytics.

However, this does not mean Uruguay is stagnant. In fact, Uruguay's close-knit football ecosystem and culture of innovation can be leveraged as advantages: it's easier to pilot new tools, implement personalized solutions, and promote collaboration between institutions, especially between academia and professional sport.

For example, Uruguay could benefit greatly from focusing on affordable, open-source technologies and applied AI research to create contextualized solutions for local clubs. Also, increasing partnerships between universities, federations, and clubs could help close the technological gap while preserving the unique cultural identity of Uruguayan football.

Chapter 3

Study Design and Methodology

3.1. Study Design

3.1.1. Interdisciplinary Collaboration with Club Nacional de Fútbol

As mentioned in the motivation section of this thesis, this project was born out of the need to understand how data is being used in Uruguayan football, a country with a renowned reputation for developing world-class players, often linked to the mythical "garra charrúa." While this national identity is admired internationally, the mechanisms behind it remain less understood.

In this context, the opportunity arose to collaborate with Club Fútbol de Football, one of Uruguay's most prestigious and historic clubs. Known for its extensive list of achievements and its strong emphasis on youth development, Nacional provided an ideal setting for this study.

Through an interdisciplinary initiative involving the Instituto de Computación (INCO) at the Facultad de Ingeniería (Fing) and the Instituto Superior de Educación Física (ISEF), a multidisciplinary project was established with Nacional's fitness coaching team, led by Wilson Espina (head of physical preparation across youth categories) and Matías de Pablo (technical staff member of the U19 squad). The aim was to evaluate and leverage the various sources of data available at the club, with a focus on injuries and player physical condition.

Data Sources Reviewed

Following a series of exploratory meetings, a formal audit was conducted of the club's current tools for data collection and performance tracking. In the following sections, we will describe main sources of data that were considered for this investigation.

GPS Tracking Data (StatsSports)

The club uses GPS devices provided by StatsSports, a leading supplier in elite sports performance tracking. These devices record high-frequency movement data (e.g., position, speed, acceleration). The data is processed via a proprietary platform with limited external access, a key constraint for the research.

Table 3.1: Schema for the Player Telemetry CSV File

Field	**Description**	`Data Type`	**Observations**
Player ID	Unique identifier for the player.	`Integer / String`	Should be the primary key for joining with player metadata tables. Consistency is key (e.g., '1023' or 'PLYR_1023').
Player Display Name	The full name of the player as it should appear in reports or dashboards.	`Text (string)`	Useful for quick analysis, but 'Player Id' should be used for all database relations.
Session ID	Unique identifier for the specific training session or match.	`Integer / String`	Links all data points from a single event. Allows for session-based analysis.
Time	The exact timestamp when the data point was recorded.	`Datetime (ISO 8601)`	Format 'YYYY-MM-DDTHH:MM:SS.sssZ' is recommended for precision and timezone handling.
Elapsed Time	Time in seconds since the beginning of the session.	`Numeric (float)`	Useful for plotting performance over time. Often more practical than absolute timestamps for session analysis.

Table 3.1 – Continued from previous page

Field	Description	Data Type	Observations
Speed (m/s)	Instantaneous speed of the player, measured in meters per second.	`Numeric (float)`	Fundamental metric for physical performance analysis. Can be used to calculate distance covered.
Instantaneous Acceleration Impulse	A measure of the change in acceleration, reflecting explosive movements like sprints or stops.	`Numeric (float)`	Often unitless or in arbitrary units (AU). Key for identifying high-intensity actions.
Heart Rate Interval	The time elapsed between two consecutive heartbeats (R-R interval), in milliseconds.	`Integer (ms)`	Raw biometric data. Can be converted to Beats Per Minute (BPM) using the formula: 'BPM = 60000 / HR_Interval'.
Latitude	The geographic Latitude coordinate from the GPS sensor.	`Numeric (float)`	WGS 84 is the standard coordinate system. Precision should be at least six decimal places.
Longitude	The geographic longitude coordinate from the GPS sensor.	`Numeric (float)`	Paired with Latitude to plot player position on a map or field. Precision should match Latitude.

Neuromuscular Status (Tool name TBD)

Neuromuscular tests are conducted periodically using force-measuring equipment, typically involving jump tests. The frequency of these tests varies depending on competition calendars and team schedules.

Table 3.2: Schema for the Neuromuscular Jump Test CSV File

Field	Description	Data Type	Observations
Name	Full name of the athlete.	`Text (string)`	Should be consistent across all tests for the same individual.
ExternalId	A unique identifier for the athlete, often from an external system.	`Integer / String`	Use this as the primary key for joining with other athlete databases.
Test Type	The specific jump protocol performed (e.g., CMJ, Squat Jump, Drop Jump).	`Text (string)`	Should be a standardized, categorical value for filtering and comparison.
Date / Time	The date and time the test was conducted.	`Date / Time`	Crucial for longitudinal tracking and monitoring trends over time.
BW [KG]	Athlete's body weight in kilograms at the time of the test.	`Numeric (float)`	Used to normalize force, power, and impulse metrics (denoted by '/ BM').
Reps	The repetition number within the set.	`Integer`	Allows for analysis of fatigue or performance changes within a single set.
Tags	Custom keywords or labels applied to the test (e.g., "Post-Game", "Rehab").	`Text (string)`	Useful for creating specific subsets of data. Often comma-separated.
Additional Load [kg]	External weight added to the athlete (e.g., via a vest or barbell).	`Numeric (float)`	Value is '0' for unloaded jumps. Important context for performance.
Jump Height (Flight Time) [cm]	The primary outcome metric, calculated from the time the athlete is in the air.	`Numeric (float)`	A key indicator of lower body explosive power.
Flight Time: Contraction Time	The ratio of time spent in the air versus time spent generating force on the ground.	`Numeric (float)`	A measure of explosive efficiency. Higher values indicate more effective force application.
RSI-modified [m/s]	Modified Reactive Strength Index. Jump Height divided by Contraction Time.	`Numeric (float)`	Quantifies the ability to generate force quickly after a countermovement.

Table 3.2 – Continued from previous page

Field	Description	Data Type	Observations
Concentric Peak Velocity [m/s]	The maximum upward (concentric) velocity achieved during the jump.	`Numeric (float)`	Indicates how fast the athlete can propel their center of mass upwards.
Concentric Peak Force [N]	The maximum upward force generated during the jump phase.	`Numeric (float)`	A measure of peak strength application.
Force at Zero Velocity [N]	The ground reaction force at the lowest point of the countermovement.	`Numeric (float)`	Represents the force at the transition point from eccentric to concentric motion.
Eccentric Deceleration RFD / BM [N/s/kg]	Rate of Force Development during the downward braking phase, normalized by body mass.	`Numeric (float)`	Measures how quickly an athlete can apply braking forces. Key for injury risk and performance.
Eccentric Peak Power / BM [W/kg]	Maximum power generated during the downward (eccentric) phase, normalized by body mass.	`Numeric (float)`	Indicates the ability to absorb and manage high eccentric loads.
Countermovement Depth [cm]	The vertical displacement of the center of mass during the countermovement.	`Numeric (float)`	Describes the jump strategy; how deep the athlete squats before jumping.

Administrative Records (Excel Spreadsheets)

Each player's personal and basic athletic information, including regular playing position, is manually recorded by the club's administrative staff. These records are not integrated with other systems and must be processed separately.

Table 3.3: Schema for the Football Players CSV File

Field	Description	Type	Comments
Player	Player's Full Name	`Text (string)`	Identifier along with document
Date of Birth	Player's Date of Birth	`Date (YYYY-MM-DD)`	Use ISO format for consistency
Identification Document	ID Number	`Text / Numeric`	No periods or hyphens
Passport	Passport Number (if applicable)	`Text (string)`	Can be "N/A"
Family Phone	Landline	`Text (string)`	Emergency contact
Cell Phone	Mobile Phone Number	`Text (string)`	Mostly their personal number
Email	Player's Email	`Text (string)`	Validate standard format
Medical Coverage	Health Institution or Insurance Company	`Text (string)`	Can be "N/A"
Education	Educational Level Attained	`Text (string)`	Current highscool grade
Hometown	City or Town of Birth	`Text (string)`	Should be standardized
Home Club	First club played for	`Text (string)`	May vary depending on availability
Home Club (%)	Economic rights of home club	`Numeric (float)`	Value between 0 and 100
Position	Usual position on the field	`Text (string)`	Use standardized categories (e.g., FIFA/UEFA)

Match Reports (PDFs provided by AUF)

The Uruguayan Football Association (AUF) supplies standardized PDF reports summarizing each match, encompassing critical data points such as team line-ups, substitutions, goal events, officiating referees, and stadium information. However, the Portable Document Format (PDF) is an inherently unstructured format designed for human readability and document fidelity, not

for direct machine processing. This presents a significant challenge for automated data extraction, as the reports lack a structured data layer that would permit conventional parsing methods.

To surmount this obstacle, a sophisticated information extraction pipeline predicated on Retrieval-Augmented Generation (RAG) was designed and implemented. This approach, originally proposed by Lewis et al., 2021 for open-domain question answering, combines the parametric knowledge of a Large Language Model (LLM) with non-parametric information retrieved from an external corpus—in this case, the content of the PDF match reports.

This RAG-based methodology was selected for its robustness against minor variations in document layout and its ability to process complex, natural language queries, offering a significant advantage over brittle, template-based extraction scripts. A comprehensive technical exposition of this data extraction pipeline, including the specific models, chunking strategies, and vectorization techniques employed, will be detailed in the subsequent chapter.

Match Video Recordings (360° Camera)

The club employs a 360-degree automatic tracking camera for home matches, offering footage that follows the ball. In away matches, where the same equipment is unavailable, a manual cameraperson is hired. This leads to inconsistencies in video angle and quality across matches.

The primary source of tactical data is derived from video recordings of each match. For home matches, the club utilizes a 360-degree automatic tracking camera system, which provides consistent, relatively high-vantage-point (4 meters) footage algorithmically centered on the ball's position. This consistency is crucial for reliable data extraction. Conversely, for away matches, the absence of this standardized equipment necessitates the use of manually operated cameras, typically resulting in significant variability in camera angle, height, zoom level, and overall video quality.

This collection of video recordings represents a rich but highly unstructured source of spatio-temporal data. To transform this raw visual information into a structured, analyzable format, a sophisticated computer vision pipeline is employed. This pipeline is tasked with processing the footage to perform object detection and tracking, enabling the extraction of key event data and the precise positional coordinates of each player and the ball throughout the match.

The notable heterogeneity between the automated home-match footage and the manually filmed away-match recordings presents a non-trivial challenge. It requires the computer vision models (mentioned in 2.3.6) to be exceptionally robust and adaptable to these inconsistencies to ensure the reliability and validity of the extracted data.

3.1.2. Population and Sample

Initially, the goal was to include in the scope data from all youth players registered at the club, ranging from the U15 squad up to the third division. A few iterations were conducted across all categories, but finally, it was concluded that technical and organizational constraints led to a narrowed focus. The primary challenge was the tool's poor usability of StatsSport, which stored most of the GPS data and required a rudimentary process to extract the information. This presented a significant barrier from the outset.

Ultimately, the U19 team during the 2024 official championship was selected as the primary sample for this investigation. This decision was based on:

- The completeness and consistency of data available for that category
- Direct communication with two key staff members: Espina and De Pablo
- The competitive context of an ongoing national tournament

This allowed for a more detailed and reliable analysis of both the raw data and its contextual interpretation.

3.1.3. Championship schedule

After establishing the objective group for our study, we sought a method to track each player's progress. The championship schedule for this category served as the default route, enabling us to gain a deeper understanding of each player's progression. Since this depends on the AUF[1], the data source was the official fixture on their webpage, with a special note that the established date at the beginning of the match could change, for any external reason, and the game could also be partially suspended, which happened in round 7, between Nacional and River. We decided to void that from the list since it affected the map.

[1] AUF: Campeonato 2024 Juveniles A Sub-19

TODO: Add an ANEXO with the chapionship

3.2. Data Collection Process

This process involves many stakeholders, with the players being the primary source of data. Therefore, it was necessary to identify all of them, create a profile for each, and include their appearances during the season. A portion of this data was already available at the beginning of the season, and additional information was being generated as the matches and the tournament progressed.

3.2.1. Equipment, Technology, and Data Collection Protocol

Before each match, the Club Nacional de Fútbol coaching staff prepared the necessary setup to gather all instance data. This involved both the use of specialized equipment and a largely manual collection process, coordinated around the routines of the coaching and medical staff.

- **GPS and Accelerometers:** Devices embedded in wearable units were distributed to players before matches and retrieved afterwards. Data was downloaded locally and uploaded post-match to the proprietary StatsSports cloud platform.
- **Neuromuscular Measurement Device:** Jump force tests were conducted off-field, typically on match days, using force plates or similar equipment. These tests were carried out both pre- and post-match for each player.
- **360° Camera and Manual Footage:** Home games were recorded with a high-quality automatic tracking camera, while away games relied on a human videographer, introducing variability in angles and quality. Video files were compiled manually and uploaded to a central archive, with timestamping that varied significantly between venues.

3.2.2. Hypothesis regarding selected variables and early challenges

A meticulous review of the available data sources revealed several unexpected issues and technical challenges. With the aim in injuries analysis and

a the goal to create a Model who could be used for predicting them, careful data validation was necessary at every stage to ensure integrity and usability. So, for the core statistical and machine learning models, the variables were classified as follows:

- Independent Variables
  - GPS-derived performance metrics: acceleration, deceleration, speed, total distance, high-intensity runs, etc.
  - Biometric data: height, weight, age, and neuromuscular force measurements.
  - Administrative data: playing position, match participation.
  - Match location: The place where the match took place
- Dependent Variable
  - Injury incidence and type: The primary objective was to predict injuries based on the previously mentioned performance metrics and biometric indicators.

3.3. Data Analysis

3.3.1. Preprocessing

One of the main technical obstacles was the lack of API access to raw data from the proprietary platform that the club uses, StatsSports. Data had to be extracted as CSV files, which, given the natural business model, the process of data extraction was no documented; we had to figure out a way of extracting raw data in the beginning. In this section, we will cover some of the processing and cleaning tasks we had to do to be able to go from data that was inside the proprietary app, up to a point where we could work with it.

Cleaning GPS Data

Each match generated over 720,000 data entries, as 10 players used vests, with a minimum of 2 hours of uptime for each device, pushing data at a rate of 10Hz. Working with raw data is usually the best approach because it allows engineers to think creatively about how to utilize the data effectively. The club had been using data for years, aggregating it to achieve specific reporting goals. But those reports usually came up directly from the proprietary software. Since

there wasn't a straightforward guide on how to extract the data, we weren't sure that the way we had found to extract it was good enough, so we needed to validate that the same reports the team was seeing using the StatsApp could be generated with the data we were seeing, to validate the theory.

One of the issues we started to see during the preliminary analysis in the beginning of this investigation was that the raw data had some outliers, as well as the natural fact that there is some delay from the moment the coaches turn on the devices and then provide them for the players to start using them during the match, and even going out to the field. For the positional outliers caused by poor satellite signal, we had to think about interpolating using surrounding timestamps and geographic locations based on the limitations of the pitch. And a natural question that could arise is that each of the fields where the team played during the championship had to be known, so when we tied each game to the actual GPS data to process and perform statistical analysis, a normalization process was needed, involving the transformation of those player positions on a 2D pitch, GPS coordinates were normalized to fit known field dimensions. A fixed (0,0) origin was used, usually at a corner flag, and other points triangulated, since in a real scenario the fields had an angle.

3.3.2. Match footage Analysis

Scene Detection Model (Video Processing)

Because 360° video footage lacked standardized timestamps, a scene detection model was built to identify the start and end of each half. This was essential to align the GPS and video data.

Key challenges:

- Inconsistent player re-entry behavior after halftime
- Lack of coordination with camera operators
- Varying video quality (especially for away matches)

Match analysis with Computer Vision

Early-stage models were tested using:

- Optical Flow
- Convolutional Neural Networks

These were meant to explore potential for future integration with richer datasets or higher-quality footage.

Comparison with Prior Studies

This research aligns with and expands upon existing literature in several areas:

- Injury prediction models using workload data
- Integration of GPS and Video

However, unique constraints in Uruguay (e.g., limited access to proprietary systems) highlight the contextual complexity of applying such techniques in under-resourced environments.

Predictive Modeling

The goal was to develop injury prediction models, using both traditional and deep learning methods:

- Random Forests
- Support Vector Machines (SVM)
- Neural Networks: MLPs, Convolutional Neural Networks (CNNs) for vision tasks

3.4. Validation and Evaluation

Cross-Validation

Cross-validation was used throughout to ensure model robustness. Additionally:

- Pre-trained player detection models were tested on club footage
- Similar techniques were applied to track the ball in matches

Chapter 4

Results

4.1. Exploratory Analysis

The analysis began with manual exploration of the raw GPS telemetry from the StatsSports devices (described in 3.1.1). Since the club's proprietary platform offered no API, we exported the data as CSV files and processed them using Jupyter notebooks. Each match's data included on average over 720,000 records, corresponding to ten players pushing data at a frequency of 10 Hz. At this step a few data pipelines where built to clean and normalize this data, handling issues like device startup delays and occasional GPS dropouts:

1. Early data points were offset by the time between powering on the vests and the match kickoff, particularly this means that there were some entries which had a GPS location outside of the pitch.
2. Some GPS fixes were erratic; we dealt with these by trimming initial time values and interpolating outlier positions within realistic pitch boundaries.

Once the data was loaded, we computed basic performance metrics (speed, distance covered, etc.) and began plotting time series to identify patterns. We found that the acceleration-related fields were particularly informative. In the telemetry schema (table 3.1) the **Instantaneous Acceleration Impulse** was defined as a measure of the change in acceleration which we could use for identifying high-intensity actions. By plotting this over each match, we saw clear peaks corresponding to sprints and rapid changes of pace. These peaks often aligned with important match events; for example, in (FIGURE XXX) a sequence of high impulses coincided with a counterattack leading to a shot. In

this way, even without advanced modeling, visual inspection of accelerations and timestamps allowed us to discover key moments in the data. These initial breakthroughs – made entirely with manual, notebook-based processing – demonstrated that the granular GPS data contained valuable insights about player effort and game context, despite the noise and volume of raw signals.

TODO: FIGURE + video capture

4.2. Dashboard Development

To make the data actionable for coaches, we developed an open-source web dashboard. The dashboard's motivation was to bridge the gap between the raw tracking data and the coaching staff's needs, centralizing multiple data sources in one interface . It provided near-real-time reports after each training session or match, giving coaches immediate feedback on loads and recovery. The dashboard architecture featured a Rust-based backend (chosen for speed and reliability) and a Next.js frontend (for a responsive interface). Building this system required several design iterations: we integrated feedback from coaches at each step, refined visualization layouts, and solved engineering challenges such as matching multiple CSV schemas and ensuring data consistency.

The dashboard's main features included:

- **Visual summaries of individual player performance** pointing to the particular match.
- **Weekly workload and neuromuscular trends**: charts of GPS load and jump-test scores over time).
- **Alerts for outliers**: automatic flags when a player's performance or load was unusually high or low).
- **Integrated match reports** (showing match-specific peaks and key events alongside video clips).

These features directly addressed the fragmented information noted in 3.1.1, where we started from different units used separate systems and the data source reviewed. In use, the dashboard was a kick starter for both to validate the raw data against what coaches already saw in proprietary tools, ensuring consistency in the reporting data, plus it served as foundation stone as a stable ground where we could build on top of. Moving forward, the dash-

board bridged data science with coaching practice: it let technical staff leverage analytics in their routine and helped close the integration gap identified in our background research.

4.3. Technical–Tactical Synchronization

Aligning the video footage with the GPS data proved challenging. As detailed in 3.3.2, our 360° match recordings lacked standardized timestamps, so we initially attempted an automated approach. A scene-detection model was built to identify the start and end of each half in the videos. In theory, this would anchor the game clock, but in practice, it was thwarted by real-world issues: players often re-entered the pitch at irregular times after halftime, and camera operators did not always coordinate with the match clock. We also experimented with player-based heuristics (for example, matching a distinctive sprint on video to a corresponding spike in GPS data), but these methods failed reliably due to occlusions and variable video quality.

TODO: MENTION THE CORELATION APPROACH

Ultimately, we resorted to manual synchronization. By comparing known events (such as kickoff, goals, or substitutions) between the video and the timestamped GPS data, we could align the streams. This labor-intensive method achieved an alignment accuracy of ±5 seconds for the two matches we completed. In particular, the Danubio match served as a case study: we noted the on-screen scoreboard changes and referee whistles to sync the half clocks. This process underscored the broader data-fusion challenge described earlier: linking the technical (accelerations and speed metrics from GPS) and tactical (from video recording) domains is complex when data sources are not naively integrated. Nonetheless, by manually matching key events, we were able to combine perspectives for those matches, enabling future combined analyses of position and actions.

The report showed that **TODO: EXTRACT and WRITE THE Analysis from DANUBIO'S MATCH**

4.4. Computer Vision Applications

We explored computer-vision (CV) methods on three of the match videos to extract spatiotemporal and tactical information. Although the footage had only 360° resolution and lacked calibration, we treated the CV experiments as feasibility studies. Consistent with approaches described in 2.3.6, we applied algorithms like optical flow and convolutional neural networks (CNNs) to track movement . For example, an optical-flow analysis highlighted regions of motion on the field (indicating team shifts or ball movement), and a CNN trained for player detection provided rough counts of players in different zones. These models were not accurate enough for precise analytics, but they did reveal overall patterns – for instance, the direction of attack or clustering of players during set pieces.

TODO: Describe de ML process, tagging players (both teams), referee, ball, goalkeepers

These early models aligned with our methodological plan (3.3.2): "Optical Flow, Convolutional Neural Networks" were tested on the match footage . The results confirmed that even low-quality video can yield some useful signals. Although not used in final metrics, the exercise validated the concept of integrating vision data: it showed that, given better cameras in the future, one could derive rich tactical features from video alone. In short, our CV work was exploratory – intended to "explore potential for future integration with richer datasets" . It demonstrated the feasibility of augmenting GPS with vision, pointing the way for more advanced tracking if higher-resolution video becomes available.

4.5. Worst Case Scenario Analysis

Inspired by existing literature on Worst Case Scenarios (WCS) in football, comparative studies were performed on:

Match peaks in individual workload

- Team performance evolution across the season
- Neuromuscular degradation post-match

TODO: ADD TABLE ANALYSIS AND QUERYs

4.6. Injury prediction

Not possible due to a lack of data, but we could fake it (?

TODO: VALIDATE WITH ANDRES BASED ON MATIAS's HIPOTHESIS and THESIS

We were unable to develop an injury-prediction model (no injuries were recorded in sufficient detail). However, we performed a preliminary analysis of one known injury risk factor: abrupt changes of direction. Biomechanical studies suggest that sudden turns (around 90°) impose high stresses on muscles and joints. Using the GPS tracks, we counted how many near-90° direction changes each player executed per match (by calculating the angle between successive movement vectors). This count acted as a crude indicator of potentially risky movements.

Although we did not have injury outcomes to correlate, the exercise was instructive. It showed that players typically performed 15–30 such turns per match, often during quick transitions or dribbling. This exploratory metric aligns with the principle that rapid accelerations/decelerations are linked to injury mechanisms. In future work, a larger dataset could test if players with higher counts are more injury-prone. For now, this study has established a pipeline for flagging high-risk actions using only GPS data, which could be integrated with clinical monitoring in the future.

4.7. The investigation itself

This work achieved significant insights despite the local constraints. Chapter 2.4 highlighted that Uruguay's football programs are generally under-resourced (smaller budgets, limited specialists, and few automated pipelines). In this context, our project demonstrates that a collaborative and open approach can make a tangible difference. By leveraging university partnerships and open-source tools, we built analytics and software that approach practices seen internationally. The scientific value lies not only in the specific findings above but also in demonstrating a practical model for performance science in Uruguay.

We moved beyond intuition to data-driven feedback and validated the hypothesis that coaches could benefit of having metrics and visualizations that mirror what top clubs achieve with proprietary systems. The dashboard and

analyses together implement the multidisciplinary vision, showing how engineering and data science can support sports practice. In summary, by delivering a usable dashboard, novel WCS analyses, and longitudinal monitoring within real-world constraints, this project provides a tangible example of elevating local practice to international standards. This interdisciplinary effort thus benefits both the scientific community (by validating techniques in a constrained setting) and the football community (by providing tools to improve player performance and care).

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APPENDICES

Appendix 1

Processed Data

ANNEXES

Annex 1

Anexo segundo

Texto de ejemplo.